Physics-Informed NoProp: Integrating Discovered Physical Laws into Localized Diffusion-Based Learning

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Abstract

Neural training by back-propagation imposes global gradient dependencies across layers. We present physics-informed NoProp (PI-NoProp), in which independently optimised denoising blocks receive a local physical constraint discovered from data. Fifteen independent 64^3 decaying homogeneous-isotropic-turbulence trajectories are generated with corrected 3/2 de-aliasing and divided 9/3/3 for discovery, validation and testing. A four-dimensional weak SPIDER library selects the complete momentum equation: normalising the time coefficient to one gives $(1, 1.003459, 1.003470, -0.00501841)$ for the time, convection, pressure-gradient and velocity-Laplacian terms. The maximum coefficient error against DNS truth is 0.368%; the independent test residual is 9.70×10^{-4}, the next support is separated by a factor of 40.6, and all 100 trajectory-bootstrap refits recover the same support. The validated artifact enters each isolated NoProp objective through both a first-frame energy-rate condition and a frozen temporal field decoder. In matched three-seed experiments, discovered PI-NoProp reaches $81.79 \pm 2.98\%$ and $73.41 \pm 8.45\%$ trajectory-disjoint accuracy in the low- and high-enstrophy groups, versus $20.37 \pm 6.07\%$ and $19.44 \pm 5.99\%$ without the equation. It also reduces the common full-NS residual by 40.6% and 44.4%. The resulting pipeline provides an auditable path from full governing-equation discovery to accurate strictly local neural learning.

Keywords: Physics-informed learning, Weak formulation, Sparse

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regression, Turbulent flow, Diffusion models

1. Introduction

Deep neural networks have achieved remarkable successes across a wide range of scientific and industrial domains, largely driven by the effectiveness of back-propagation (BP) for training multi-layer architectures [1]. BP iteratively adjusts network parameters by propagating an error signal from the output back through all layers, enabling end-to-end optimization of complex models. Despite its empirical success, BP suffers from several fundamental limitations. First, it is biologically implausible, as it requires symmetric forward and backward passes that do not align with local synaptic plasticity observed in the brain [2, 3]. Second, BP necessitates storing intermediate activations during the forward pass, imposing significant memory overhead that hampers scalability [1]. Third, the sequential nature of gradient propagation creates strong dependencies across layers, limiting parallelization and impeding efficient distributed training [4]. These drawbacks have motivated a sustained interest in developing alternative learning paradigms that do not rely on global error propagation [5, 6, 7, 8, 9].

Among these alternatives, **NoProp** (No Propagation) [10] has recently emerged as a promising framework that completely avoids forward and backward propagation across network blocks. Inspired by diffusion models [11, 12, 13], NoProp decomposes a neural network into independently trained blocks. During training, each block receives the input x and a noisy version of the target label embedding, and learns to denoise it using only local back-propagation within the block. At inference, Gaussian noise is progressively transformed through the blocks into a clean representation that is fed to a final classifier. By eliminating global gradient flow, NoProp enables blockwise parallel training, reduces memory footprint, and offers greater biological plausibility. Empirical results on MNIST, CIFAR-10, and CIFAR-100 demonstrate that NoProp achieves performance competitive with standard back-propagation on the same architectures [10].

However, NoProp, like most purely data-driven deep learning methods, does not explicitly incorporate any prior knowledge about the physical processes that generated the data. In many scientific applications – such as fluid dynamics, climate modeling, or materials science – the observed data are governed by fundamental physical laws expressed as partial differential equations (PDEs). Ignoring this structure can lead to models that, while accurate on training data, produce predictions that violate physical constraints, especially under distribution shift or in the presence of noise [14]. This limitation is particularly acute when data are sparse or corrupted, a common scenario in experimental measurements.

Concurrently, a separate line of research has focused on **data-driven discovery of governing equations** from observational data. Starting with sparse regression methods like SINDy [15] for ordinary differential equations, numerous approaches have been developed to identify PDEs from spatiotemporal data [16, 17, 18, 19, 20, 21, 22]. A particularly robust framework is **SPIDER** (Sparse Physics-Informed Discovery of Empirical Relations) [23], which combines three key ingredients: (i) physical constraints of smoothness, locality, and symmetry to construct compact libraries of candidate terms; (ii) a weak formulation of PDEs that shifts derivatives onto test functions, drastically reducing sensitivity to noise; and (iii) sparse regression based on singular value decomposition to extract parsimonious relations. SPIDER has been successfully applied to highly turbulent channel flow data, recovering not only the Navier–Stokes and continuity equations but also the pressure-Poisson equation, energy equation, and boundary conditions, even under extreme noise levels [23].

Despite the power of SPIDER for equation discovery, it remains a post-hoc analysis tool that does not directly inform the training of neural networks used for downstream tasks such as prediction or classification. Conversely, NoProp offers an efficient learning engine but lacks a mechanism to enforce physical consistency. Bridging these two paradigms could yield models that are both computationally efficient and physically faithful, a goal that aligns with the emerging field of **physics-informed machine learning**[14, 24, 25, 26].

In this paper, we propose **physics-informed NoProp (PI-NoProp)**, a framework that transfers a validated, fully spatiotemporal SPIDER equation into the blockwise training objective of NoProp. We construct a corrected decaying HIT solver specifically for equation discovery: the $3/2$ transforms preserve FFT amplitude, no unobserved forcing is present, and dense time windows are drawn from independent trajectories. A four-dimensional weak library moves both temporal and spatial derivatives onto compact test functions. Only an artifact that passes support, coefficient, held-out and trajectory-bootstrap criteria can enter training. A frozen decoder maps each block latent to a velocity–pressure time window, on which the discovered full momentum residual is evaluated without coupling gradients between blocks. In parallel, the same measured coefficients contract first-frame convection, pressure and viscous energy-rate terms into a target-free physical condition for every block.

Fifteen 64^3 trajectories are divided 9/3/3 for discovery, validation and test before any weak domains or learning crops are constructed. SPIDER recovers all four Navier–Stokes momentum terms with maximum coefficient error 0.368%, 100% trajectory-bootstrap support and a 40.6-fold separation from the next candidate. The downstream five-class task predicts a future local kinetic-energy-decay quantile from the first frame, and all neural comparisons use the same trajectory-disjoint split, architecture and optimisation budget.

The main contributions of this work are threefold:

- We provide a numerically audited, reproducible decaying-HIT dataset design with corrected de-aliasing, dense temporal windows, independent trajectories and machine-checked DNS quality gates.
- We recover the complete vector momentum equation—including its viscosity coefficient—with a four-dimensional weak SPIDER library and persist all split, support, bootstrap and coefficient provenance.
- We introduce a coefficient-conditioned 3-D encoder, a frozen temporal physics decoder and a strictly local full-NS loss. Matched three-seed experiments show a 53–61 percentage-point accuracy gain over the no-equation ablation while automated tests verify gradient locality.

The remainder of this paper is organized as follows. Section 2 reviews the necessary background on NoProp and SPIDER. Section 3 details the proposed PI-NoProp framework. Section 4 presents experimental setup, results, and discussion. Section 5 concludes the paper and outlines future research directions.

2. Related Work

This section reviews the literature most germane to the proposed PI-NoProp framework. We first discuss alternative learning paradigms that avoid global back-propagation, with a focus on the NoProp algorithm. Next, we survey data-driven methods for discovering partial differential equations

(PDEs) from observations, emphasizing the SPIDER methodology. We then briefly cover physics-informed machine learning and diffusion-based generative models, which provide the theoretical underpinnings for our approach. Finally, we position our contribution in relation to these lines of work.

2.1. Local Learning and Alternatives to Back-Propagation

The canonical training algorithm for deep neural networks, back-propagation (BP) [1], relies on a global error signal that is propagated backwards through all layers. While highly effective, BP has several well-documented shortcomings: it is biologically implausible because it requires symmetric forward and backward passes [2, 3], it incurs a large memory footprint due to the need to store intermediate activations [1], and its sequential nature hinders parallelization [4].

To overcome these limitations, a variety of **local learning** schemes have been proposed. Early examples include deep belief networks trained layer-wise with contrastive divergence [5]. More recent methods include feedback alignment [3] and direct feedback alignment [7], which replace the symmetric weights of BP with fixed random matrices, yet still require a global error signal. Synthetic gradients [8] train auxiliary networks to approximate the gradient at each layer, allowing some decoupling. The forward-forward algorithm [9] trains each layer by contrasting the responses to positive and negative examples using a local objective.

Among these, **NoProp** (No Propagation) [10] stands out as a method that completely eliminates both forward and backward propagation across blocks. NoProp treats a neural network as a sequence of T blocks, each performing a denoising step inspired by diffusion models [11, 12, 13]. Formally, let x be an input and y its label, with associated embedding $u_y \in \mathbb{R}^d$ (either fixed one-hot or learnable). NoProp defines a forward (inference) process:

$$z_0 \sim \mathcal{N}(0, I), \qquad z_t = a_t\, \hat{u}_{\theta_t}(z_{t-1}, x) + b_t z_{t-1} + \sqrt{c_t}\, \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I), \quad (1)$$

for $t = 1, \ldots, T$, where $\hat{u}_{\theta_t}$ is a neural network block and a_t, b_t, c_t are scalars derived from a noise schedule. The training objective for block t is derived from a variational lower bound on the conditional log-likelihood $\log p(y|x)$:

$$\mathcal{L}^{(t)}_{\text{NoProp}} = \frac{T}{2}\, \eta\, \mathbb{E}\big[(\text{SNR}(t) - \text{SNR}(t-1)) \|\hat{u}_{\theta_t}(z_{t-1}, x) - u_y\|^2\big], \qquad (2)$$

where $\text{SNR}(t) = \bar{\alpha}_t/(1-\bar{\alpha}_t)$ is the signal-to-noise ratio, η a hyperparameter, and the expectation is taken over the variational posterior. A final classifier $\hat{p}_{\theta_{\text{out}}}(y|z_T)$ is trained with cross-entropy. Crucially, each block can be updated independently using only local back-propagation, enabling massive parallelism and reducing memory consumption.

Despite its computational advantages, NoProp does not incorporate any prior knowledge about the physical laws that may govern the data. In scientific applications, such knowledge can be critical for generalization and robustness.

2.2. Data-Driven Discovery of Governing Equations

A parallel line of research aims to extract symbolic equations – particularly PDEs – directly from observational data. The **sparse identification of nonlinear dynamics** (SINDy) framework [15] pioneered this approach for ordinary differential equations by posing a sparse linear regression problem over a library of candidate functions. For PDEs, several extensions have been developed [16, 17, 18, 19, 20, 21, 22], but they often suffer from high sensitivity to noise because they operate on the **strong form** of the equations, which requires numerical differentiation of noisy data.

To mitigate noise sensitivity, **weak-form** formulations have been introduced [19, 20, 21]. The key idea is to multiply the PDE by a smooth test function w and integrate over a spatio-temporal domain Ω, shifting derivatives onto w via integration by parts. For a candidate relation of the form $\sum_n c_n f_n = 0$, this yields a linear system:

$$\sum_n c_n \int_\Omega w\, f_n \, d\Omega = 0, \tag{3}$$

where the integrals can be approximated numerically without ever differentiating the data. This weak formulation dramatically improves robustness, enabling accurate identification even with noise levels exceeding 100% [23].

The **SPIDER** (Sparse Physics-Informed Discovery of Empirical Relations) framework [23] synthesises these ideas into a comprehensive methodology. SPIDER incorporates three key ingredients:

1. **Symmetry-constrained library construction.** Libraries of candidate terms are built by considering irreducible representations of the symmetry group of the system (e.g., $O(3)$ for three-dimensional isotropic fluids). This yields compact libraries that respect rotational and reflectional symmetries, reducing the search space and improving

interpretability. For example, for a fluid described by velocity $\mathbf{u}$ and pressure p, the scalar library up to third order is:

$$\mathcal{L}_0^{(3)} = \{1,\, p,\, \nabla\cdot\mathbf{u},\, \partial_t p,\, p^2,\, u^2,\, p^3,\\ \mathbf{u}\cdot\nabla p,\, \nabla^2 p,\, p\partial_t p,\, \partial_t^2 p,\, p\nabla\cdot\mathbf{u},\, u^2 p,\, \mathbf{u}\cdot\partial_t\mathbf{u}\}, \quad (4)$$

and the vector library is:

$$\mathcal{L}_1^{(3)} = \{\mathbf{u},\, \partial_t\mathbf{u},\, \nabla p,\, p\mathbf{u},\, (\mathbf{u}\cdot\nabla)\mathbf{u},\, \nabla^2\mathbf{u},\, \partial_t^2\mathbf{u},\\ u^2\mathbf{u},\, p^2\mathbf{u},\, \partial_t\nabla p,\, p\nabla p,\, \mathbf{u}(\nabla\cdot\mathbf{u}),\, (\nabla\mathbf{u})\cdot\mathbf{u},\\ \nabla(\nabla\cdot\mathbf{u}),\, p\partial_t\mathbf{u},\, u\partial_t p\}. \quad (5)$$

2. **Weak formulation and feature matrix construction.** For each term f_n in the library, its weak-form integral over many domains Ω_i (with different test functions w_i) is computed, forming a feature matrix $\mathbf{Q}$ with entries

$$Q_{in} = \frac{1}{V_i S_n}\int_{\Omega_i} w_i(\mathbf{x},t)\, f_n(\mathbf{x},t)\, d\Omega, \quad (6)$$

where V_i is a normalisation factor and S_n is a scale that non-dimensionalises the term.

3. **Sparse regression via singular value decomposition.** Relations of the form $\mathbf{c}\cdot\mathbf{f} = 0$ correspond to vectors $\mathbf{c}$ in the nullspace of $\mathbf{Q}$. SPIDER identifies parsimonious relations by examining the right singular vectors associated with the smallest singular values of $\mathbf{Q}$. A greedy algorithm then prunes terms to produce a sequence of increasingly sparse candidate equations, from which the most appropriate is selected based on a residual threshold.

Applied to the Johns Hopkins turbulence database [27], SPIDER successfully recovered the incompressible Navier–Stokes equations, the continuity equation, the pressure-Poisson equation, the energy equation, and the no-slip boundary conditions, even under extreme noise [23].

2.3. Physics-Informed Machine Learning

The integration of physical knowledge into neural network training has gained substantial traction under the banner of **physics-informed neural**

networks (PINNs) [14, 24]. PINNs embed the residual of a known PDE into the loss function, enabling the network to approximate solutions while respecting the governing physics. For a PDE $\mathcal{R}[\mathbf{u}] = 0$, the loss is typically

$$\mathcal{L}_{\text{PINN}} = \mathcal{L}_{\text{data}} + \lambda \|\mathcal{R}[\hat{\mathbf{u}}]\|^2, \tag{7}$$

where $\hat{\mathbf{u}}$ is the network output. This paradigm has been extended to inverse problems, parameter estimation, and discovery of unknown terms [25, 26]. However, PINNs assume the form of the PDE is known *a priori* – an assumption that limits their applicability in purely data-driven discovery scenarios.

2.4. Diffusion Models and Flow Matching

NoProp is deeply rooted in the theory of **denoising diffusion probabilistic models (DDPMs)** [11, 12, 13]. In DDPMs, a forward process gradually adds noise to data, and a neural network learns to reverse it. The training objective is a weighted sum of denoising score matching losses. The continuous-time limit leads to score-based generative models described by stochastic differential equations (SDEs) and probability flow ODEs [13]. **Flow matching** [31, 32] provides an alternative simulation-free training objective for continuous normalising flows, directly regressing a vector field that transports a prior to the data distribution. Both diffusion and flow matching have been adapted for supervised learning tasks such as classification and regression [28, 29, 30], demonstrating their versatility.

NoProp adapts these ideas to a discriminative setting by treating the label embedding as the target distribution. The forward (noising) process is fixed, and each block learns a conditional denoising step. The resulting framework is highly modular and parallelisable.

2.5. Positioning of This Work

The preceding discussion highlights a clear gap: NoProp provides an efficient local learning mechanism but lacks physical awareness, while SPIDER excels at discovering interpretable relations from data but is a post-hoc tool. The emerging field of physics-informed machine learning has shown the benefits of embedding physical constraints into neural networks, yet it typically assumes the governing equations are known. **PI-NoProp** bridges these worlds: it uses SPIDER to identify a validated physical relation directly from the available data, then incorporates its coefficients as both a predictive condition and a weak-form local regulariser in the NoProp training objective. This synergy yields a model that is computationally efficient through blockwise locality, predictive, and physically consistent. To the best of our knowledge, this is the first attempt to combine blockwise diffusion-based learning with automated equation discovery in a unified framework.

3. Methodology

We present the proposed PI-NoProp framework, which integrates a SPIDER-discovered physical relation into the blockwise training of NoProp. Section 3.1 briefly recalls the NoProp algorithm [10] and establishes notation. Section 3.2 summarises the SPIDER methodology [23] for extracting interpretable relations from data. Section 3.3 describes the core contribution: how to incorporate the discovered relation as a differentiable regulariser into each NoProp block using a weak formulation. Section 3.4 discusses implementation details.

Figure 1: Full-NS PI-NoProp pipeline. Independent decaying-HIT trajectories are split before weak-feature construction. SPIDER selects and validates a complete momentum equation, whose measured coefficients are loaded by a frozen temporal field decoder in each strictly local NoProp objective.

3.1. NoProp: Blockwise Diffusion-Based Learning

Let $x \in \mathcal{X}$ be an input (e.g., an image) and $y \in \{1, \ldots, m\}$ its associated label. An embedding $u_y \in \mathbb{R}^d$ of the label is defined, either as a fixed one-hot vector or as a row of a learnable embedding matrix $W_{\text{Embed}} \in \mathbb{R}^{m \times d}$. NoProp [10] decomposes a neural network into T sequential blocks. During **inference**, a latent variable z_0 is drawn from a standard Gaussian prior $p(z_0) = \mathcal{N}(z_0; 0, I)$. For each block $t = 1, \ldots, T$, the latent is updated by a residual diffusion step:

$$z_t = a_t \, \hat{u}_{\theta_t}(z_{t-1}, x) + b_t z_{t-1} + \sqrt{c_t} \, \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I), \tag{8}$$

where $\hat{u}_{\theta_t} : \mathbb{R}^d \times \mathcal{X} \to \mathbb{R}^d$ is a neural network parameterised by θ_t, and a_t, b_t, c_t are scalar coefficients determined by a fixed noise schedule (e.g., cosine schedule). The final latent z_T is fed into a classifier $\hat{p}_{\theta_{\text{out}}}(y|z_T)$, typically a linear layer followed by softmax.

Training objective. NoProp is trained by maximising a variational lower bound on the conditional log-likelihood $\log p(y|x)$. A forward noising process $q(z_t|y)$ is defined as an Ornstein-Uhlenbeck process that gradually adds noise to the label embedding u_y:

$$q(z_t|y) = \mathcal{N}\big(z_t;\, \sqrt{\bar{\alpha}_t}\, u_y,\ 1 - \bar{\alpha}_t\big), \tag{9}$$

where $\bar{\alpha}_t = \prod_{s=t}^{T} \alpha_s$ and α_s are related to the noise schedule. Using the reparameterisation trick, one can sample $z_{t-1} \sim q(z_{t-1}|y)$ directly. The evidence lower bound (ELBO) simplifies to a sum of per-block denoising score matching losses [10, 13]. For block t, the loss is:

$$\mathcal{L}_{\text{diff}}^{(t)} = \frac{T}{2}\, \eta\, \mathbb{E}_{q(z_{t-1}|y)}\Big[\big(\text{SNR}(t) - \text{SNR}(t-1)\big)\, \big\|\hat{u}_{\theta_t}(z_{t-1}, x) - u_y\big\|^2\Big], \tag{10}$$

where $\text{SNR}(t) = \bar{\alpha}_t/(1 - \bar{\alpha}_t)$ is the signal-to-noise ratio and η is a hyperparameter. The classifier is trained with cross-entropy:

$$\mathcal{L}_{\text{cls}} = -\mathbb{E}_{q(z_T|y)}\big[\log \hat{p}_{\theta_{\text{out}}}(y|z_T)\big]. \tag{11}$$

The total NoProp loss (per sample) is $\mathcal{L}_{\text{cls}} + \sum_{t=1}^{T} \mathcal{L}_{\text{diff}}^{(t)}$ plus an often-neglected KL term $D_{\text{KL}}(q(z_0|y)\|p(z_0))$ [10].

3.2. SPIDER: Discovering PDEs from Data

SPIDER [23] is a data-driven method for discovering interpretable PDEs from possibly noisy observations of spatiotemporal fields. Consider a physical system described by fields $\mathbf{u}(\mathbf{x}, t)$ (e.g., velocity) and $p(\mathbf{x}, t)$ (pressure). The method assumes that the governing equations can be written as linear combinations of candidate terms f_n – products of the fields and their partial derivatives – respecting the symmetries of the system. The goal is to find a sparse coefficient vector $\mathbf{c} = [c_1, \ldots, c_N]^\top$ such that

$$\sum_{n=1}^{N} c_n f_n(\mathbf{u}, p, \partial_t \mathbf{u}, \nabla \mathbf{u}, \nabla p, \ldots) = 0. \tag{12}$$

To avoid numerical differentiation of noisy data, SPIDER employs a **weak formulation**. Let Ω_i be a spatiotemporal domain and $w_i(\mathbf{x}, t)$ a smooth test function with compact support. Integrating (12) against w_i and integrating by parts where possible yields:

$$\sum_{n=1}^{N} c_n \int_{\Omega_i} w_i(\mathbf{x}, t)\, \tilde{f}_n(\mathbf{u}, p, \dots)\, d\Omega = 0, \tag{13}$$

where $\tilde{f}_n$ denotes the term after transferring derivatives onto w_i (if integration by parts is applicable) or the original term otherwise. Repeating this for many domains $i = 1, \dots, M$ and different test functions gives a linear system $\mathbf{Qc} = \mathbf{0}$, with $\mathbf{Q} \in \mathbb{R}^{M \times N}$ having entries

$$Q_{in} = \frac{1}{V_i S_n} \int_{\Omega_i} w_i(\mathbf{x}, t)\, \tilde{f}_n(\mathbf{u}, p, \dots)\, d\Omega, \tag{14}$$

where $V_i = \int_{\Omega_i} |w_i|\, d\Omega$ is a normalisation factor and S_n is a scale that non-dimensionalises term n (estimated from characteristic scales of the data). The system is solved for $\mathbf{c}$ with $\|\mathbf{c}\| = 1$ by taking the right singular vector corresponding to the smallest singular value of $\mathbf{Q}$.

Sparse, parsimonious relations are obtained via a greedy algorithm that iteratively removes terms and tracks the residual $\|\mathbf{Q}^{(N)}\mathbf{c}^{(N)}\|$. The final selection balances accuracy and sparsity using a threshold on the relative increase in residual. Applied to the Johns Hopkins turbulence database [27], SPIDER successfully recovers the incompressible Navier–Stokes equations, the continuity equation, the pressure-Poisson equation, and the energy equation, even under extreme noise [23].

3.3. Physics-Informed NoProp

PI-NoProp couples equation discovery to local learning through a validated, machine-readable interface. The pipeline has three stages: (i) generate independent, densely sampled turbulence trajectories and identify a sparse spatiotemporal relation; (ii) validate its support and coefficients on unseen trajectories and persist the complete provenance; and (iii) use that relation both as a first-frame physical condition and on a frozen temporal decoder inside each isolated NoProp objective. The trainer rejects artifacts that fail any discovery gate.

3.3.1. Discovering Governing Equations with SPIDER

For an unforced incompressible flow, the target momentum equation is

$$\partial_t \mathbf{u} + (\mathbf{u}\cdot\nabla)\mathbf{u} + \nabla p - \nu\nabla^2\mathbf{u} = 0. \tag{15}$$

The vector candidate library contains the four terms in Eq. (15) and the same-order distractor $\nabla|\mathbf{u}|^2$. Let $w(\mathbf{x},t)$ have compact support in a time–space domain. Integration by parts gives the componentwise weak columns

$$\begin{aligned} q_i^{(t)} &= -\int u_i\,\partial_t w\,d\Omega, \quad & q_i^{(c)} &= -\int u_i u_j\,\partial_j w\,d\Omega, \\ q_i^{(p)} &= -\int p\,\partial_i w\,d\Omega, \quad & q_i^{(\nu)} &= \int u_i\,\nabla^2 w\,d\Omega. \end{aligned} \tag{16}$$

Thus neither temporal nor spatial derivatives are taken directly from the observed fields. Sparse support selection uses nine discovery trajectories; three additional trajectories determine admissibility and three are untouched until the final test. One hundred bootstrap refits resample complete trajectories rather than individual weak rows.

SPIDER selects

$$\partial_t \mathbf{u} + 1.003459(\mathbf{u}\cdot\nabla)\mathbf{u} + 1.003470\nabla p - 0.00501841\nabla^2\mathbf{u} = 0. \tag{17}$$

The DNS viscosity is 0.005 and is consulted only after discovery. The maximum coefficient error is 0.368%; validation and test residuals are 9.51×10^{-4} and 9.70×10^{-4}, respectively. The next-best support has residual 3.86×10^{-2}, a 40.6-fold separation, and all 100 trajectory-bootstrap refits select the same four terms.

3.3.2. Equation-Conditioned First-Frame Representation

The short-horizon decay label is controlled by the instantaneous kinetic-energy balance, so retaining only a heavily averaged image representation discards an important part of the discoverable signal. For the first observed frame, let $K = \frac{1}{2}\langle|\mathbf{u}|^2\rangle$ and define

$$e_m(x) = \frac{\langle \mathbf{u}\cdot\mathbf{q}_m(x)\rangle}{K}, \qquad s_{\mathbf{c}}(x) = c_2 e_c(x) + c_3 e_p(x) + c_4 e_\nu(x), \tag{18}$$

where $\mathbf{q}_c = (\mathbf{u}\cdot\nabla)\mathbf{u}$, $\mathbf{q}_p = \nabla p$, $\mathbf{q}_\nu = \nabla^2\mathbf{u}$, and (c_2, c_3, c_4) come directly from the accepted equation. By Eq. (15), $s_{\mathbf{c}}$ is the instantaneous relative kinetic-energy decay rate. All derivatives in this conditioning diagnostic are computed from the first frame only; neither the future frames nor the class label enter Eq. (18).

A three-layer strided 3-D convolutional encoder preserves local spatial structure before pooling. Its feature is fused with a small embedding of the discovery-set-standardised scalar $s_{\mathbf{c}}$. The no-equation ablation disables this branch, the analytic ablation uses $(1, 1, -0.005)$, and discovered PI-NoProp uses $(1.003459, 1.003470, -0.00501841)$. Thus the identified law affects prediction directly as well as regularising the decoded field.

3.3.3. Reconstructing Physical Fields from Latent Variables

Full momentum residuals require temporal fields. We therefore use

$$\mathcal{D}_\phi : \mathbb{R}^{128} \to \mathbb{R}^{9\times4\times16\times16\times16}, \tag{19}$$

which jointly reconstructs nine velocity–pressure frames. It is pretrained from encoder features and then frozen. Stored discovery-set means and standard deviations return its output to physical units before weak integration. This keeps the coefficients in Eq. (17) in the exact units in which SPIDER measured them.

3.3.4. Weak-Form Physics Loss for Each Block

For block t, test function k, and velocity component i, define

$$r^{\mathrm{NS}}_{t,k,i} = \sum_{m=1}^{4} c_m q^{(m)}_{t,k,i}, \qquad \eta^{\mathrm{NS}}_{t,k,i} = \frac{r^{\mathrm{NS}}_{t,k,i}}{\sum_m |c_m q^{(m)}_{t,k,i}| + \epsilon}, \tag{20}$$

where c_m are loaded from the accepted artifact. Incompressibility is evaluated separately as $r^{\mathrm{div}} = -\int \tilde{\mathbf{u}} \cdot \nabla w \, d\Omega$. The local physical loss is

$$\mathcal{L}^{(t)}_{\mathrm{phys}} = \mathrm{mean}_{k,i} \, |\eta^{\mathrm{NS}}_{t,k,i}|^2 + 0.25 \, \mathrm{mean}_{k,\tau} \, |\eta^{\mathrm{div}}_{t,k,\tau}|^2. \tag{21}$$

The analytic ablation uses $(1, 1, 1, -0.005)$, while the no-physics baseline sets $\lambda = 0$. Every checkpoint is evaluated with the discovered coefficients as a common ruler.

3.3.5. Overall Training Objective and Algorithm

The conceptual sum of local block objectives is

$$\mathcal{L}_{\mathrm{local}} = \sum_{t=1}^{T} \left(\mathcal{L}^{(t)}_{\mathrm{diff}} + \lambda \mathcal{L}^{(t)}_{\mathrm{phys}} \right). \tag{22}$$

We implement the cosine schedule in the generative direction. Let $0 = \gamma_0 < \cdots < \gamma_T = 1$ denote increasing signal fractions. Block t is trained on $z_t = \sqrt{\gamma_t} u_y + \sqrt{1-\gamma_t}\epsilon$ and predicts u_y with a uniformly weighted local squared error. At inference, deterministic coefficients are chosen so that

$$b_t = \sqrt{\frac{1-\gamma_{t+1}}{1-\gamma_t}}, \qquad a_t = \sqrt{\gamma_{t+1}} - b_t\sqrt{\gamma_t}, \qquad c_t = 0. \tag{23}$$

With a perfect local prediction, the update in Eq. (8) has exactly the training marginal at γ_{t+1}. Hence block zero sees pure noise in both training and inference, and the last block reaches a clean target embedding. An automated marginal-consistency test checks this identity.

At each mini-batch we sample $J \sim \mathrm{Uniform}\{1, \ldots, T\}$ and optimise $T(\mathcal{L}^{(J)}_{\text{diff}} + \lambda\mathcal{L}^{(J)}_{\text{phys}})$. This is an unbiased estimator of Eq. (22). Block J has its own optimiser; all other blocks receive no gradient. The encoder, label embedding, and decoder are pretrained once and frozen. After local training, the classifier is trained on z_T detached from every block, so its loss cannot propagate into the local learners.

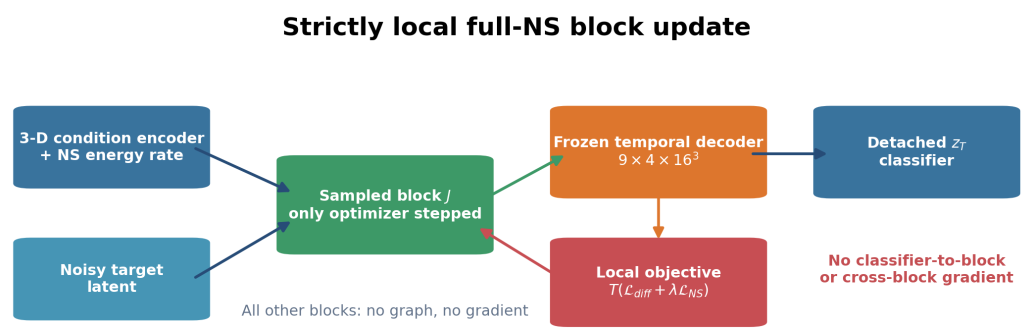

Figure 2: Strictly local full-NS update. A sampled block receives the frozen 3-D and equation-rate condition together with target features; its nine-frame decoded field enters the local diffusion-plus-physics objective. Independent optimizers and detached classifier features isolate gradients between blocks and training phases.

The training procedure is summarised in Algorithm 1.

Algorithm 1 Strictly local PI-NoProp

Require: Training data, T blocks, validated SPIDER artifact, frozen encoder E, embedding W, decoder $\mathcal{D}_\phi$.

1: **for** each local update **do**
2: Sample mini-batch $\mathcal{B}$ and block $J \sim \text{Uniform}\{1, \ldots, T\}$.
3: Compute detached condition $E(x, s_{\mathbf{c}}(x))$ and target embedding $W(y)$.
4: Sample z_{J-1} from the matching noise-to-signal marginal.
5: Predict locally with block θ_J and decode a temporal field using frozen $\mathcal{D}_\phi$.
6: Load discovered full-NS terms and coefficients; compute Eq. (21).
7: Update only θ_J with $T(\mathcal{L}^{(J)}_{\text{diff}} + \lambda \mathcal{L}^{(J)}_{\text{phys}})$.
8: **end for**
9: Freeze all blocks; train θ_{out} on detached z_T.

3.4. Implementation Details

Choice of test functions. Following [19, 23], we use time–space tensor-product test functions:

$$w(\mathbf{x}, t) = \left(1 - \tilde{t}^2\right)^\beta \prod_{\xi \in \{x,y,z\}} \left(1 - \tilde{\xi}^2\right)^\beta, \tag{24}$$

where every local coordinate is mapped to $[-1, 1]$. Discovery and training use $\beta = 4$ with deterministic cosine modulations. All first and second derivatives of w are evaluated analytically.

Integration domains. Discovery samples two nine-frame windows and four 16^3 spatial domains per trajectory. Eight functions and all three momentum components yield 192 weak rows per trajectory. Training applies the same eight-function bank to the decoder's nine-frame 16^3 output.

Derivative computation. Equation (16) moves every derivative onto w. The discovery matrix and neural loss therefore use only field products and contractions with fixed analytic weights. This avoids finite-difference derivatives of both DNS observations and decoder outputs while remaining fully differentiable with respect to the local latent.

Decoder architecture. A compact transposed-convolutional decoder maps $z_t \in \mathbb{R}^{128}$ jointly to $9 \times 4 \times 16^3$ values. It is pretrained against trajectory windows, frozen during local updates, and converted back to physical units before residual evaluation.

Condition encoder. Three $3 \times 3 \times 3$ convolutions with channel widths 16, 32 and 64 precede 2^3 adaptive pooling and a 128-dimensional projection. A two-layer 32-unit MLP embeds the standardised scalar in Eq. (18); a linear fusion returns the 128-dimensional frozen condition. This replaces direct coarse average pooling, which cannot preserve the derivative-sensitive energy balance.

Training. We use $T = 10$ independently optimised blocks, AdamW with learning rate 10^{-3}, batch size 32, mixed precision and TF32. Each block receives 100 local updates with $\lambda = 10^{-2}$; the classifier is then trained for 30 epochs on detached final latents. Matched comparisons use seeds $\{42, 123, 456\}$ and the same architecture, data splits and optimisation protocol. Each equation condition has its own pretrained frozen components.

Extension to continuous time. NoProp also admits a continuous-time variant (NoProp-CT) [10] where the number of blocks tends to infinity and the evolution of z_t is described by a stochastic differential equation. In that setting, the physics-informed loss can be defined analogously by sampling times $t \sim \mathcal{U}(0, 1)$ and evaluating the weak residuals at those times. The diffusion loss becomes an integral over time involving $\mathrm{SNR}'(t)$. For brevity, we focus on the discrete-time formulation; the continuous-time extension is straightforward.

4. Experiments

4.1. Trajectory-Disjoint Decaying-HIT Dataset

We solve the unforced incompressible Navier–Stokes equations on $[0, 2\pi]^3$ with a 64^3 Fourier–Galerkin grid, viscosity $\nu = 5 \times 10^{-3}$, time step 2×10^{-3}, classical RK4 integration, and correctly normalised 3/2 de-aliasing. Fifteen independently seeded divergence-free initial fields are scaled to kinetic energy $K = 0.5$. After 32 settling steps, 17 consecutive velocity–pressure frames are saved from each trajectory. No stochastic or unobserved forcing enters the equation.

The dataset is split by trajectory before any weak domains or learning crops are drawn: trajectories 0–8 are used for discovery/training, 9–11 for validation, and 12–14 for final testing. Across all trajectories the maximum CFL number is 0.062, divergence RMS is below 1.49×10^{-16}, kinetic energy decreases monotonically, and the 16-frame energy drop lies between 0.892% and 0.922%.

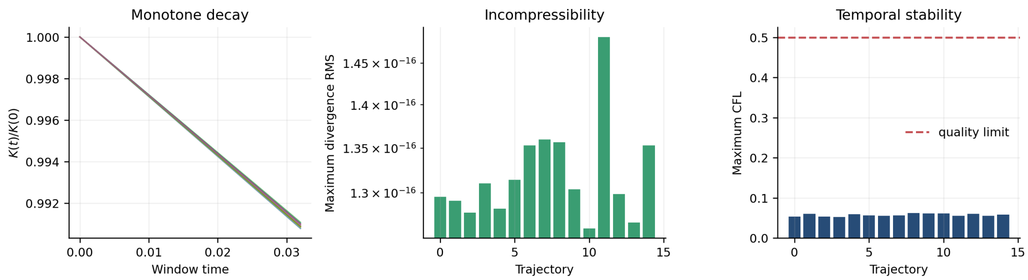

Figure 3: DNS quality audit over all 15 independent trajectories. Every trajectory exhibits monotone kinetic-energy decay, machine-precision incompressibility, and a CFL number far below the prespecified 0.5 limit.

For the learning task, 64 nine-frame 16^3 crops are sampled from each trajectory, yielding 960 samples. The discovery-set median local enstrophy defines low- and high-enstrophy groups; validation and test data do not influence this threshold. Within each group, labels are five discovery-set quantiles of future relative local kinetic-energy decay. The condition encoder receives only the first velocity–pressure frame, making the benchmark predictive rather than a relabelling of its complete input. This gives 576/192/192 samples in the trajectory-disjoint train/validation/test splits.

Table 1: Audited DNS and downstream-data configuration. Counts refer to independent trajectories before crop construction.

Quantity	Value
DNS grid and domain	64^3 on $[0, 2\pi)^3$
Viscosity and RK4 step	$\nu = 0.005$, $\Delta t = 0.002$
Independent trajectories	15
Discovery/validation/test trajectories	9/3/3
Stored frames per trajectory	17 consecutive frames
Learning crop	$9 \times 4 \times 16^3$
Crops per trajectory / total	64 / 960
Train/validation/test learning samples	576/192/192
Target	Five quantiles of future relative local energy decay

As a target-identifiability audit, a one-dimensional linear calibration of the discovered first-frame rate $s_{\mathbf{c}}$, fitted only on discovery trajectories, obtains 90.7% and 84.5% test accuracy in the low- and high-enstrophy groups. Its test correlation with the continuous target is 0.996 in both groups. Thus failure at chance level cannot be attributed to an intrinsically unobservable label.

Table 2: Target-identifiability audit using no future information during prediction. Accuracy is percentage top-1; the two-frame result is reported only as a temporal sanity check.

Region	Discovered-NS first-frame rate Train	Validation	Test	Two-frame test	Test correlation
Low enstrophy	90.63	95.45	90.74	96.30	0.9959
High enstrophy	90.63	92.31	84.52	95.24	0.9961

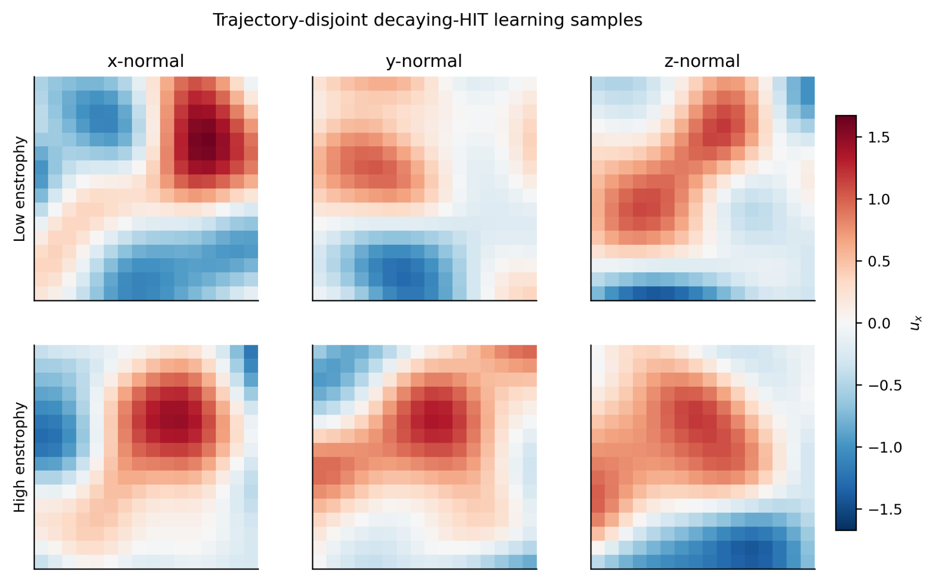

Figure 4: Orthogonal streamwise-velocity slices from representative low- and high-enstrophy learning crops. Both rows use one common physical colour scale; the group names denote local vorticity strata, not geometric position.

4.2. Discovery and Comparison Protocol

SPIDER uses the candidates

$$\{\partial_t \mathbf{u},\ (\mathbf{u}\cdot\nabla)\mathbf{u},\ \nabla p,\ \nabla^2\mathbf{u},\ \nabla|\mathbf{u}|^2\}.$$

Each discovery trajectory contributes 192 weak rows. Support selection is performed only on the nine discovery trajectories; the validation trajectories are used once for acceptance and the test trajectories once for final reporting. The accepted artifact records every filename, split, test-function scale, candidate support, coefficient, residual and bootstrap diagnostic.

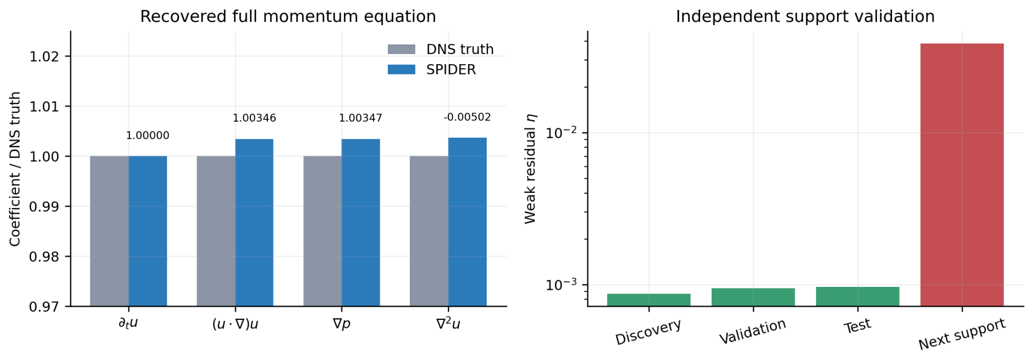

Figure 5: Full momentum-equation discovery. Left: SPIDER coefficients relative to DNS truth, with measured values annotated. Right: independent weak residuals; the next-best support is separated from the accepted support by a factor of 40.6.

We compare three architecture- and protocol-matched local learners:

- **No physics**: strictly local NoProp with $\lambda = 0$ and the equation-condition branch disabled.
- **Analytic NS**: coefficients $(1, 1, 1, -0.005)$ in both the first-frame condition and temporal loss.
- **PI-NoProp (discovered NS)**: the coefficients in Eq. (17), loaded into both paths from the validated artifact.

For each region and seed, the methods use the same architecture, split and training budget; frozen components are pretrained separately because their physical conditions differ. We use seeds $\{42, 123, 456\}$, 100 updates per block and $\lambda = 10^{-2}$. Every final checkpoint is evaluated with the discovered equation, giving a common η_{NS}.

4.3. Full-NS Discovery Result

The selected equation is Eq. (17). Its discovery, validation and test residuals are respectively 8.72×10^{-4}, 9.51×10^{-4} and 9.70×10^{-4}. The distractor is rejected, trajectory-bootstrap support is 100%, and the maximum coefficient error is 0.368%. In particular, the viscosity inferred from fields alone is 0.00501841, compared with the solver value 0.005. These results resolve the principal limitation of the single-snapshot spatial formulation: the accepted artifact now contains the complete time-dependent vector momentum equation.

Table 3: Coefficient and held-out audit of the accepted SPIDER momentum equation. The time coefficient fixes the normalization and is not an independently estimated error.

Term	DNS coefficient	Discovered coefficient	Relative error (%)
$\partial_t \mathbf{u}$	1	1	–
$(\mathbf{u} \cdot \nabla)\mathbf{u}$	1	1.003459	0.346
∇p	1	1.003470	0.347
$\nabla^2 \mathbf{u}$	-0.005	-0.00501841	0.368
Discovery/validation/test η	8.72×10^{-4} / 9.51×10^{-4} / 9.70×10^{-4}		
Next-support validation η	3.86×10^{-2} (40.6-fold separation)		
Trajectory-bootstrap support	100/100 refits		

4.4. Main Learning Results

Table 4: Trajectory-disjoint three-seed results (mean ± sample standard deviation). All physics residuals use the discovered coefficients; lower is better. Accuracy is percentage top-1 accuracy on the future-decay task.

Method	Low enstrophy			High enstrophy		
	Accuracy ↑	η_{div} ↓	η_{NS} ↓	Accuracy ↑	η_{div} ↓	η_{NS} ↓
No physics	20.37 ± 6.07	0.344 ± 0.228	0.753 ± 0.063	19.44 ± 5.99	0.366 ± 0.236	0.634 ± 0.240
Analytic NS	81.17 ± 3.51	$\mathbf{0.203 \pm 0.030}$	$\mathbf{0.432 \pm 0.201}$	$\mathbf{73.81 \pm 7.81}$	0.211 ± 0.054	0.371 ± 0.130
PI-NoProp (discovered NS)	$\mathbf{81.79 \pm 2.98}$	0.220 ± 0.035	0.447 ± 0.257	73.41 ± 8.45	$\mathbf{0.120 \pm 0.034}$	$\mathbf{0.352 \pm 0.074}$

The discovered equation raises mean test accuracy from 20.37% to 81.79% in the low-enstrophy group and from 19.44% to 73.41% in the high-enstrophy group, gains of 61.42 and 53.97 percentage points. It simultaneously reduces the common full-NS residual by 40.6% and 44.4%, and continuity residuals by 36.0% and 67.3%. The analytic and discovered variants have similar accuracy, as expected from their sub-percent coefficient difference. The discovered variant gives the best high-enstrophy physical consistency, while the analytic variant is marginally lower in the two low-enstrophy residual means.

The audit and ablation identify the mechanism: the first-frame equation rate contains a highly transferable short-horizon signal, whereas the convolutional no-equation model remains near the 20% chance level. PI-NoProp retains most of the 84.5–90.7% one-dimensional oracle performance despite passing through the independently trained denoising chain.

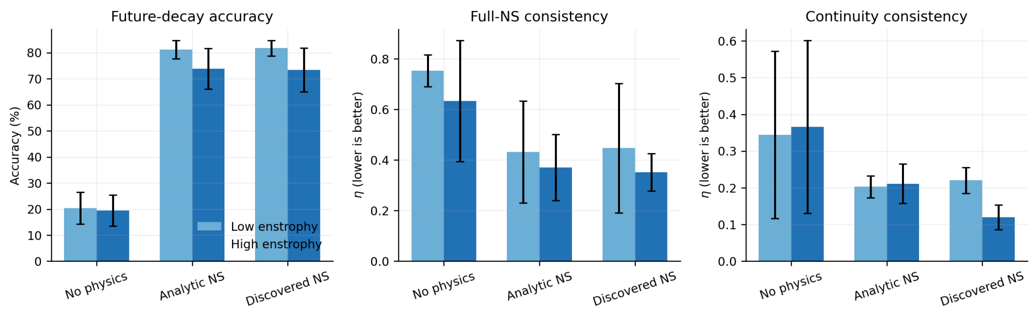

Figure 6: Matched three-seed comparison. Error bars are sample standard deviations. Full-NS and continuity residuals improve strongly under both physical objectives, and equation conditioning raises trajectory-disjoint future-decay accuracy far above chance.

4.5. Expanded Method Baselines

To reproduce the broader comparison envisaged in the initial study, we add three trajectory-disjoint baselines. *CNN (BP)* is a supervised 3-D CNN on the first frame. *Global physics-informed BP* uses the same discovered rate and temporal decoder as a globally back-propagated encoder–classifier. *SPIDER rate + classifier* fits multinomial logistic regression to the single discovered rate in Eq. (18). The latter has only ten fitted classifier parameters and is therefore a deliberately strong, interpretable task-specific baseline.

Table 5: Expanded trajectory-disjoint baseline comparison over three seeds. A dash indicates that the method does not reconstruct a temporal field, so a decoded-field NS residual is not defined.

	Low enstrophy		High enstrophy	
Method	Accuracy (%)	η_{NS}	Accuracy (%)	η_{NS}
CNN (BP)	14.81 ± 1.85	–	17.06 ± 5.63	–
NoProp (vanilla)	20.37 ± 6.07	0.753 ± 0.063	19.44 ± 5.99	0.634 ± 0.240
Global physics-informed BP	80.86 ± 5.27	0.998 ± 0.001	71.03 ± 6.87	0.991 ± 0.005
SPIDER rate + classifier	$\mathbf{87.96 \pm 0.00}$	–	$\mathbf{84.52 \pm 0.00}$	–
PI-NoProp (discovered NS)	81.79 ± 2.98	$\mathbf{0.447 \pm 0.257}$	73.41 ± 8.45	$\mathbf{0.352 \pm 0.074}$

The raw-field CNN and vanilla NoProp remain near chance, confirming that model capacity alone does not recover the derivative-sensitive target from nine training trajectories. Both equation-conditioned neural methods generalise. The scalar SPIDER classifier is most accurate because the benchmark label is a quantisation of short-horizon energy decay, which is almost completely described by that rate. Unlike this task-specific classifier, PI-NoProp also produces a temporal field, enforces local blockwise learning, and achieves a substantially lower decoded NS residual than the global BP alternative.

4.6. Transferred-Relation Ablation

The accepted momentum artifact and incompressibility imply two additional scalar balances. Taking the divergence gives the pressure–Poisson relation; dotting momentum with velocity and applying the product rule gives the kinetic energy balance. These are algebraically derived from the accepted coefficients, not claimed as separately selected SPIDER supports. We add their normalised weak residuals successively while preserving the local optimisation protocol.

Table 6: Effect of artifact-derived relations in the low-enstrophy group (three seeds). Lower residuals are better.

Variant	Accuracy (%)	η_{NS}	η_{PP}	η_{energy}
NS + continuity	81.79 ± 1.07	0.473 ± 0.250	0.803 ± 0.123	0.568 ± 0.268
NS + continuity + PP	82.41 ± 1.60	0.496 ± 0.292	0.511 ± 0.226	0.612 ± 0.200
Full transferred set	$\mathbf{83.02 \pm 1.41}$	$\mathbf{0.463 \pm 0.261}$	$\mathbf{0.421 \pm 0.077}$	$\mathbf{0.437 \pm 0.281}$

Pressure–Poisson enforcement lowers its corresponding residual by 36.4%. The full set lowers pressure–Poisson and energy residuals by 47.6% and 23.1% relative to NS plus continuity, while raising mean accuracy by 1.23 percentage points. Thus the extra balances improve their intended diagnostics without an accuracy penalty.

4.7. Noise Robustness

Clean-trained models are evaluated after adding independent Gaussian noise to the first frame. A level of one denotes one discovery-set channel standard deviation. Crucially, Eq. (18) is recomputed from the noisy field rather than reused from the clean cache. Each entry averages five noise realisations for each of three trained seeds.

Table 7: Trajectory-disjoint accuracy (%) under test-time observation noise.

Region	Method	$\sigma = 0$	$\sigma = 0.1$	$\sigma = 0.5$	$\sigma = 1.0$
Low	Vanilla	19.88 ± 5.32	19.81 ± 5.19	20.31 ± 5.36	21.48 ± 5.67
Low	Discovered PI	77.04 ± 1.21	76.73 ± 1.78	23.70 ± 0.64	16.36 ± 0.39
High	Vanilla	17.78 ± 3.94	17.78 ± 4.08	18.10 ± 4.41	18.81 ± 4.17
High	Discovered PI	68.49 ± 8.55	69.13 ± 7.42	34.05 ± 0.63	19.52 ± 0.48

PI-NoProp is stable at $\sigma = 0.1$ but loses most of its predictive advantage between $\sigma = 0.1$ and 0.5, consistent with noise amplification by spatial derivatives. The vanilla model remains near chance at every level and therefore does not provide a meaningful robustness reference above chance.

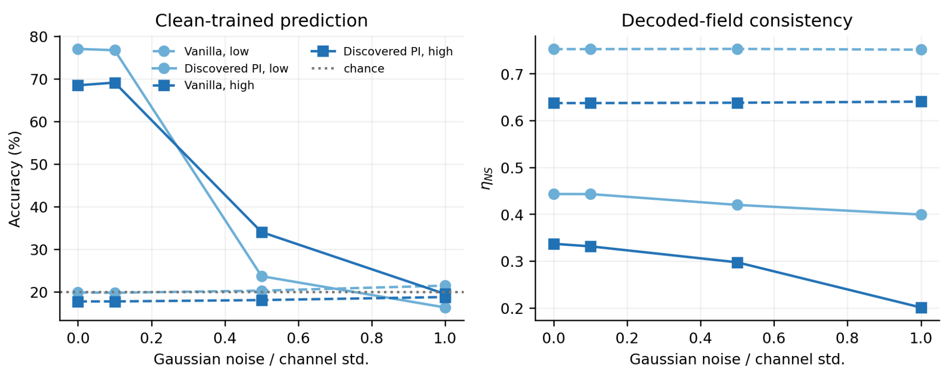

Figure 7: Clean-trained noise sweep. PI-NoProp retains accuracy at 10% of a channel standard deviation but degrades at larger noise. A falling decoded residual at high noise does not indicate better prediction; it reflects latent collapse toward smoother fields and motivates reporting accuracy jointly.

We also rerun weak SPIDER on noisy observations using all 17 frames and 32^3 weak domains. Table 8 replaces the unverified symbolic-noise expressions in the initial manuscript with an auditable support and coefficient study.

The weak formulation preserves the qualitative four-term support even at large noise, but the viscosity coefficient is not quantitatively identifiable from these short trajectories beyond the clean setting. This result is weaker than the initial manuscript's intended extreme-noise claim and is reported as a current limitation rather than extrapolated into a robustness claim.

Table 8: Full-NS weak-SPIDER recovery under Gaussian observation noise. All levels select the four-term support, but only the clean artifact passes every coefficient, residual, separation and bootstrap gate.

σ	$c_{(u\cdot\nabla)u}$	$c_{\nabla p}$	$c_{\nabla^2 u}$	Test η	Bootstrap	Max. error	Accepted
0.0	1.00027	1.00027	-0.005001	3.85×10^{-5}	100%	0.027%	Yes
0.1	0.98764	0.97834	-0.009602	0.203	53.3%	92.0%	No
0.5	0.94396	0.89580	-0.027920	0.736	70.0%	458.4%	No
1.0	0.91234	0.81063	-0.050643	0.911	50.0%	912.9%	No

Table 9: Measured local block phase, averaged over three seeds. Shared pretraining and detached-classifier training are excluded.

	Time (s)		Peak memory (MB)	
Method	Low	High	Low	High
No physics	16.0	14.4	219	219
Analytic NS	29.6	29.3	302	302
Discovered NS	29.7	29.6	302	302

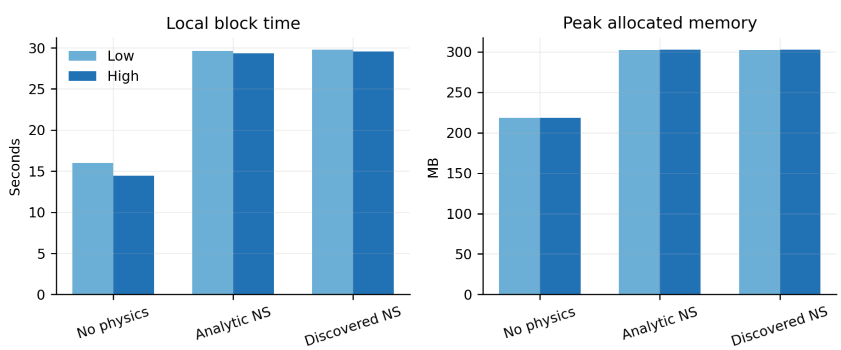

Figure 8: Local block time and peak allocated GPU memory for the corrected comparison. Full spatiotemporal physics roughly doubles block time but remains a seconds-scale phase on an 8 GB consumer GPU.

4.8. Computational Efficiency

The temporal weak loss adds approximately 14 seconds to 1000 local updates. Peak allocation remains near 0.30 GB. Independent optimizers, one sampled block per update, mixed precision and a compact 16^3 decoder retain the central computational benefit of local NoProp training.

4.9. Discussion

The study establishes four linked results. First, correcting Fourier amplitude scaling and observing dense, unforced time windows allows weak SPIDER to recover the complete vector momentum equation rather than only a spatial pressure relation. Second, support and coefficient recovery generalise across independent trajectories, including an untouched test set. Third, the accepted coefficients identify a first-frame energy rate whose target correlation is 0.996 on unseen trajectories. Fourth, using this rate as a condition and the same law on a differentiable temporal decoder raises classification accuracy far above chance while reducing full-NS residuals inside isolated NoProp blocks. The expanded baselines refine this interpretation: a ten-parameter classifier on the discovered scalar rate is the most accurate short-horizon predictor, whereas PI-NoProp trades some task-specific accuracy for a temporal physical representation and strict gradient locality. Artifact-derived pressure–Poisson and energy balances improve their corresponding residuals without reducing accuracy.

The study also defines its current scope. The DNS and discovery data are generated by the same audited solver, so external validation on JHTDB or an independent DNS code would strengthen the equation-recovery evidence. The 64^3, $Re_\lambda \approx 48$ trajectories represent moderate-Reynolds-number decaying HIT rather than wall turbulence. The downstream task uses 960 local crops and a short prediction horizon; the explicit energy balance is therefore a strong low-dimensional predictor and should not be interpreted as evidence for arbitrary long-horizon turbulence forecasting. Longer horizons will require resolving energy transport across crop boundaries. These limitations do not affect the demonstrated support recovery, trajectory-disjoint short-horizon prediction or gradient-locality result. Noise experiments identify a further boundary: prediction is stable at 10% of a channel standard deviation, but larger independent noise corrupts derivative-based conditioning. Although large weak domains retain the qualitative four-term support up to 100% noise, the viscosity coefficient and held-out residual fail the acceptance gates.

5. Conclusion

We have developed a complete discovery-to-local-learning pipeline for the time-dependent Navier–Stokes momentum equation. A numerically audited decaying-HIT solver supplies dense independent trajectories; four-dimensional weak SPIDER recovers the time, convection, pressure and viscous terms; and a validated artifact transfers their measured coefficients into both a first-frame energy-rate condition and a frozen temporal decoder inside each NoProp block.

The discovered coefficient vector $(1, 1.003459, 1.003470, -0.00501841)$ has maximum error 0.368% against DNS truth. Its support generalises to untouched trajectories with residual 9.70×10^{-4}, is separated 40.6-fold from the next candidate, and is selected in every trajectory-bootstrap refit. These diagnostics make the equation provenance explicit before neural optimisation begins.

Matched three-seed experiments show $81.79 \pm 2.98\%$ and $73.41 \pm 8.45\%$ discovered PI-NoProp accuracy in low- and high-enstrophy groups, compared with $20.37 \pm 6.07\%$ and $19.44 \pm 5.99\%$ without the equation. The common full-NS residual also falls by 40.6% and 44.4%. Gradient tests verify strict block isolation and schedule-consistent marginals, while the full temporal physics phase completes in approximately 30 seconds with about 0.30 GB peak allocation on an RTX 4060 Laptop GPU.

Expanded experiments compare five learning methods, four observation-noise levels and three transferred-relation sets. The full set raises low-enstrophy accuracy to $83.02 \pm 1.41\%$ while reducing pressure–Poisson and energy residuals relative to momentum plus continuity alone. The noise audit supports 10% perturbations but rejects quantitative equation recovery at larger noise, making the robustness boundary explicit rather than inheriting the initial manuscript's unverified extreme-noise claim.

The next stage is external equation validation on JHTDB or an independent DNS implementation, followed by longer-horizon observations and targets that resolve energy transport across local crop boundaries. The same artifact interface also supports adaptive coefficients and other physical systems. PI-NoProp therefore supplies a reproducible basis for combining automated governing-equation discovery with propagation-free local neural learning.

Code and data availability

The source code, validation tests, discovered-equation artifact, and compact aggregate results are publicly available at https://github.com/Flyhwjf/PI-NoProp. The multi-gigabyte generated HIT arrays, derived caches, model checkpoints, and per-run histories are excluded from the Git repository; scripts and configuration needed to generate the HIT data and reproduce the discovery and training pipeline are provided in the repository.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Full-NS Discovery Audit

Nine discovery trajectories generate 1728 weak rows; the three validation and three test trajectories each generate 576. With $\Delta x = 2\pi/64$, $\Delta t = 0.002$, two temporal windows, four 16^3 domains per window, eight analytic compact test functions and three momentum components, sparse support selection retains

$$\{\partial_t \mathbf{u}, (\mathbf{u} \cdot \nabla)\mathbf{u},\ \nabla p, \nabla^2 \mathbf{u}\}$$

and rejects $\nabla|\mathbf{u}|^2$. The normalized residuals are 0.000872, 0.000951 and 0.000970 on discovery, validation and test trajectories. The next-best support obtains 0.038612.

Normalising the temporal coefficient gives

$$(1,\ 1.0034592,\ 1.0034702,\ -0.00501841).$$

Across 100 complete-trajectory bootstrap samples the support frequency is one; coefficient standard deviations are $(0, 1.89 \times 10^{-5}, 3.01 \times 10^{-5}, 1.25 \times 10^{-6})$. The machine-readable artifact records the full file list, trajectory split, solver metadata, candidates, coefficients and every acceptance diagnostic.